Supplementary Material: VarDictRust: A Memory-Bounded, Oracle-Equivalent Rust Consolidation of the VarDict Variant-Calling Pipeline for Deep Whole-Genome and Ultra-Low-VAF Sequencing

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Engine Release: VarDictRust v1.0.6 (commit 2a8e9a3 , engine fingerprint 5865a2a0b24a8c17 , performance fingerprint c490c36317b8f25b)

Reference Oracles: VarDictJava v1.8.3 (0886261) and vardictcpp ae93fe39dc (Wolff 2026)

Cloud Benchmark Host: GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM, 4x375 GB NVMe RAID0) (1.5 TB NVMe RAID0 scratch; exploratory staging also verified on c3d-standard-32)

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1. Supplementary Methods: Exact Command-Line Invocations Across All 60 Benchmark Cells

All 60 verification cells (59 cloud compute cells plus 1 certified local-to-cloud evidence bundle evidence_certified) were executed on GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM, 4x375 GB NVMe RAID0) (Ubuntu 24.04 LTS , Linux kernel 6.8 , 1.5 TB 4-way NVMe RAID0 mounted at /var/tmp/vardict-paper). Every timed execution was wrapped in /usr/bin/time -v across n = 3 independent technical replicates (Stage 2 , Stage 3 , Stage 4) or guarded by timeout 10800 (3 hours, exit code rc=124 on timeout; rc=3 on JVM OutOfMemoryError).

1.0 Manuscript Figure and Table to 60-Cell Provenance Crosswalk

To ensure complete machine-traceable auditability while keeping the main manuscript prose accessible to general readers, the table below maps every main-text Figure (Figure 1 – Figure 8) and Table (Table 1 – Table 8) to its underlying cell_id directories in benchmarks/gcp-study-v106/ (Table S1):

- **Figure 1 & Table 1 (Architecture, Operational Modes, and 87-Arm Equivalence Sweep):** s1_ceiling_amplicon , s1_ceiling_hg008 , s1_gate7_ref_multiregion , s1_h1a_simple_tiles , s2_tierp_mode_parity_sweep , s3_rna_splice_parity_n3 , s4_multimodal_and_cram_twin , evidence_certified , and the 12 sweep_* cells (sweep_4targets_full , sweep_R_chr1_full , sweep_hg002_chr20_exome , sweep_multimodal_and_cram_twin , sweep_rna_hg002_52gb , sweep_span_chr1_f0 , sweep_tierp_13modes , sweep_tierp_SRR15006540 , sweep_tierp_SRR8657348 , sweep_tierp_somatic_76_86 , sweep_ultradeep_mrd_stress , sweep_wgs_mrd_matrix).
- **Figure 2 & Table 2 (Wolff et al. AML Cohort, Field Callers, and Contiguous -R chr1):** s2_tierp_v2_SRR15006375_th1 , s2_tierp_v2_SRR15006375_th8 , s2_tierp_v2_SRR15006376_th1 , s2_tierp_v2_SRR15006376_th8 , s2_tierp_v2_SRR15006386_th1 , s2_tierp_v2_SRR15006386_th8 , s2_tierp_v2_SRR15006540_th1 , s2_tierp_v2_SRR15006540_th8 , s2_tierp_v2_SRR8657348_th1 , s2_tierp_v2_SRR8657348_th8 , s2_tierp_v2_somatic_76_86_th1 , s2_tierp_v2_somatic_76_86_th8 , s2_table3_R_chr1_java_200g_n3 , s2_table3_R_chr1_rust_and_cpp , and s2_fig1_fig3_field_callers .
- **Figure 3 & Table 3 (Germline/Somatic Truth Concordance and QUAL Operating Points):** s2_fig2_hg002_chr20_truth , s4_tier3_truth_and_qual_sweep , s5_grch38_field_callers_truth , and evidence_certified .
- **Figure 4 & Table 4 (Multicore Scaling, Chromosome 1 Span-to-RSS Invariance, and Node Packing):** s4_perf_budget_4targets_n3 , s4_scale_exome_3way_th1 , s4_scale_exome_3way_th2 , s4_scale_exome_3way_th4 , s4_scale_exome_3way_th8 , s4_scale_exome_3way_th16 , s4_scale_exome_3way_th32 , s4_span_chr1_3way_th8 , s4_span_chr1_3way_th16 , s4_span_chr1_3way_th32 , s4_span_chr1_f0 , and s4_jobs_per_node_2x4c_n3 .
- **Figure 5 & Table 5 (Full-Genome WGS Un-Chunked Contigs vs Pre-Chunked Callable Intervals):** s5_wgs_hg002_full_contigs_3way_th32 , s5_wgs_hg002_full_giab_3way_th32 , s5_wgs_hg008_full_contigs_3way_th32 , and s5_wgs_hg008_full_giab_3way_th32 .
- **Figure 6 & Table 6 (10-Chemistry Chr20 Panel, -f 0.0 Noise Floor, and 70-Cell GA4GH v3.5 Matrix):** s5_cross_platform_short_read_chr20_panel and s6_ga4gh_v35_stratification_matrix .

- **`Figure 7` & `Table 7` (Next-Generation `GIAB v5.0Q T2T-Q100`, `CMRG v1.00`, and `Mosaic v1.10`):** `s6_hg002_v50q_cmrg_mosaic_eval`.
- **`Figure 8` & `Table 8` (11-Caller `HG008 v0.4`, Low-VAF Titration, Supply Chain, and `vardictcpp` Audit):** `s6_hg008_modern_ml_and_commercial_callers`, `s6_low_vaf_titration_truth_curve`, and `s5_h4_maintenance_security_matrix`.

1.1 Stage 1: Ceiling, Multi-Region Boundary, and Tile-Invariance Cells (`s1_*`)

Stage 1 comprises `4` cells (`s1_ceiling_amplicon`, `s1_ceiling_hg008`, `s1_gate7_ref_multiregion`, and `s1_h1a_simple_tiles`):

```
# Cell 1: s1_ceiling_amplicon (EGFR/MRE11a amplicon -a 10:0.95, 2 variants)
vardictrust -G hg19.fa -f 0.01 -N SAMPLE -b amplicon.bam \
  -a 10:0.95 --fisher -c 1 -S 2 -E 3 -g 4 --vcf amplicon.bed > rust_amp.vcf

# Cell 2: s1_ceiling_hg008 (NIST GIAB HG008 somatic slice, 18,426 variants)
vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N "HG008-T|HG008-N" \
  -b "HG008-T.bam|HG008-N.bam" -th 8 --fisher -z -c 1 -S 2 -E 3 -g 4 \
  --vcf hg008_slice.bed > rust_hg008.vcf

# Cell 3: s1_gate7_ref_multiregion (HG002 chr20 multi-region boundary gate, 2,572 variants)
bash harness/compare.sh GRCh38.fa hg002_chr20.bam chr20_multiregion.bed HG002

# Cell 4: s1_h1a_simple_tiles (Tile-size invariance: 500,000, 1,000,000, 2,000,000, 10,000,000 bp -> 5,090 variants)
for TILE in 500000 1000000 2000000 10000000; do
  VARDICTRUST_TILE_BP="${TILE}" vardictrust -G GRCh38.fa -f 0.01 -N HCC1395 \
    -b HCC1395.bam -th 8 --fisher -c 1 -S 2 -E 3 -g 4 --vcf seqc2_slice.bed > "tile_${TILE}.vcf"
done
```

1.2 Stage 2: Tier-P AML Replication, 13-Mode Sweep, Field Callers, and `-R chr1` (`s2_*`)

Stage 2 comprises `17` cells replicating and extending Wolff et al. (2026) [11]:

- **12 Tier-P v2 AML Panel Cells (`n = 3` technical replicates each at `-th 1` and `-th 8`):** `s2_tierp_v2_SRR15006375_th1`, `s2_tierp_v2_SRR15006375_th8`, `s2_tierp_v2_SRR15006376_th1`, `s2_tierp_v2_SRR15006376_th8`, `s2_tierp_v2_SRR15006386_th1`, `s2_tierp_v2_SRR15006386_th8`, `s2_tierp_v2_SRR15006540_th1`, `s2_tierp_v2_SRR15006540_th8`, `s2_tierp_v2_SRR8657348_th1`, `s2_tierp_v2_SRR8657348_th8`, `s2_tierp_v2_somatic_76_86_th1`, and `s2_tierp_v2_somatic_76_86_th8`.
- **Contiguous Chromosome 1 (`-R chr1:1-249250621`, `249,250,621` bp on `SRR8657348`):** `s2_table3_R_chr1_java_200g_n3` (`JAVA_OPTS="-Xmx4g"` OOM `rc=3` at `31.08` s; `JAVA_OPTS="-Xmx200g"` completes in `501.83` s at `106.47` GiB RSS) and `s2_table3_R_chr1_rust_and_cpp` (`VarDictRust` `9.75` s, `318.6` MB RSS vs `vardictcpp` `105.89` s, `10.58` GiB RSS).
- **13-Mode Functional Sweep & Field Callers:** `s2_tierp_mode_parity_sweep` (`13/13` modes), `s2_fig1_fig3_field_callers` (`FreeBayes`, `LoFreq`, `Mutect2`, `Octopus`, `Strelka2`), and `s2_fig2_hg002_chr20_truth` (`8,233` candidate variants).

```
# Tier-P single-sample & paired somatic replication (s2_tierp_v2_*):
/usr/bin/time -v vardictrust -G hg19.fa -f 0.01 -N SRR8657348 \
  -b SRR8657348.bam -th 8 --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf SRR8657348.bed > SRR8657348.vcf

# Table 3 contiguous chromosome 1 (-R chr1:1-249250621 on SRR8657348, s2_table3_R_chr1_*):
/usr/bin/time -v vardictrust -G hg19.fa -f 0.01 -N SRR8657348 \
  -b SRR8657348.bam -th 8 -R chr1:1-249250621 > rust_chr1.var
JAVA_OPTS="-Xmx200g" /usr/bin/time -v VarDict -G hg19.fa -f 0.01 -N SRR8657348 \
  -b SRR8657348.bam -th 8 -R chr1:1-249250621 > java_chr1.var
```

1.3 Stage 3: Full-Sample 51.8 GiB RNA-Seq Splice Calling (`s3_rna_splice_parity_n3`)

Cell `s3_rna_splice_parity_n3` (`n = 3` technical replicates) evaluates `-i` RNA splice-junction mode on the `51.8` GiB HG002 GM24385 RNA-seq alignment (`12,668` `.var` and `.splice.tsv` records, `4.94` s vs `23.31` s, `4.72` × speedup, `13.6` × RSS reduction):

```
/usr/bin/time -v vardictrust -G GRCh38.fa -f 0.01 -N HG002_RNA \
  -b HG002_GM24385_RNA.bam -i -th 8 -c 1 -S 2 -E 3 -g 4 gencode_exons.bed > rust_rna.var
```

1.4 Stage 4: Whole-Exome Scaling, Chromosome 1 Span-to-RSS, Node Packing, and `QUAL` ROC (`s4_*`)

Stage 4 comprises `15` cells (`s4_perf_budget_4targets_n3`, `s4_scale_exome_3way_th1`, `s4_scale_exome_3way_th2`, `s4_scale_exome_3way_th4`, `s4_scale_exome_3way_th8`, `s4_scale_exome_3way_th16`, `s4_scale_exome_3way_th32`, `s4_span_chr1_3way_th8`, `s4_span_chr1_3way_th16`, `s4_span_chr1_3way_th32`, `s4_span_chr1_f0`, `s4_jobs_per_node_2x4c_n3`, `s4_multimodal_and_cram_twin`, `s4_tier3_truth_and_qual_sweep`, and `evidence_certified`):

```
# Whole-exome scaling across 1, 2, 4, 8, 16, 32 threads (s4_scale_exome_3way_th1..th32):
for TH in 1 2 4 8 16 32; do
  /usr/bin/time -v vardictrust -G GRCh38.fa -f 0.01 -N "HCC1395|HCC1395BL" \
    -b "HCC1395.bam|HCC1395BL.bam" -th "${TH}" --fisher -z -c 1 -S 2 -E 3 -g 4 \
    --vcf seqc2_exome.bed > "seqc2_th${TH}.vcf"
done

# Chromosome 1 span-to-RSS invariance across 1M, 10M, 100M, 249M bp (s4_span_chr1_3way_th8..th32):
for SPAN in 1000000 10000000 100000000 248956422; do
  /usr/bin/time -v vardictrust -G GRCh38.fa -f 0.01 -N HG002 \
    -b HG002_chr1.bam -th 32 -R "chr1:1-${SPAN}" > "chr1_${SPAN}.var"
done

# All-positions (-f 0) pileup stress on chr1 10 Mbp (s4_span_chr1_f0) and 2x4c node packing (s4_jobs_per_node_2x4c_n3):
/usr/bin/time -v vardictrust -G GRCh38.fa -f 0 -N HG002 -b HG002_chr1.bam -th 8 -R chr1:10000001-20000000 > chr1_f0.var
```

1.5 Option C: Comprehensive 87-Arm `.var` and `.vcf` Parity Sweep (`sweep_*`)

The `12` Option C cells (`sweep_4targets_full` , `sweep_R_chr1_full` , `sweep_hg002_chr20_exome` , `sweep_multimodal_and_cram_twin` , `sweep_rna_hg002_52gb` , `sweep_span_chr1_f0` , `sweep_tierp_13modes` , `sweep_tierp_SRR15006540` , `sweep_tierp_SRR8657348` , `sweep_tierp_somatic_76_86` , `sweep_ultradeep_mrd_stress` , and `sweep_wgs_mrd_matrix`) validate `.var` and `.vcf` equivalence across `87` comparative arms (`84/87` `MATCH` , `96.55%` , `Table S2`):

```
# Ultra-deep 45000x ctDNA MRD stress (-f 0.0005 and -f 0 with --fisher, sweep_ultradeep_mrd_stress):
python3 harness/compare_var.py golden/ultradeep_f0005.var rust_ultradeep_f0005.var --eps 1e-6
# WGS & exome MRD matrix (-f 0.001 and -f 0, sweep_wgs_mrd_matrix):
python3 harness/compare_vcf.py golden/hg005_300x_mrd.vcf rust_hg005_300x_mrd.vcf --eps 1e-6
```

1.6 Stage 5: Full-Genome WGS (Arm A vs Arm B), 10-Platform Chr20 Panel, and Supply Chain (`s5_*`)

Stage 5 comprises `7` cells (`s5_wgs_hg002_full_contigs_3way_th32` , `s5_wgs_hg002_full_giab_3way_th32` , `s5_wgs_hg008_full_contigs_3way_th32` , `s5_wgs_hg008_full_giab_3way_th32` , `s5_grch38_field_callers_truth` , `s5_cross_platform_short_read_chr20_panel` , and `s5_h4_maintenance_security_matrix`):

```
# Arm A: Un-chunked full chromosomes (chr1..chr22, s5_wgs_hg002_full_contigs_3way_th32 & s5_wgs_hg008_full_contigs_3way_th32):
timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N HG002 \
  -b HG002_2x250_131GB.bam -th 32 --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf grch38_22autosomes_full.bed > hg002_arm_a.vcf

timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N "HG008-T|HG008-N" \
  -b "HG008-T_195x.bam|HG008-N_169x.bam" -th 32 -U --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf grch38_22autosomes_full.bed >
hg008_arm_a.vcf

# Arm B: Pre-chunked <=50 kbp GIAB callable intervals (s5_wgs_hg002_full_giab_3way_th32 & s5_wgs_hg008_full_giab_3way_th32):
timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N HG002 \
  -b HG002_2x250_131GB.bam -th 32 --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf HG002_GRCh38_v421_50kb.bed > hg002_arm_b.vcf

timeout 10800 /usr/bin/time -v vardictrust -G GRCh38_GIABv3.fasta -f 0.01 -N "HG008-T|HG008-N" \
  -b "HG008-T_195x.bam|HG008-N_169x.bam" -th 32 -U --fisher -z -c 1 -S 2 -E 3 -g 4 --vcf HG008_GRCh38_v0.3_50kb.bed >
hg008_arm_b.vcf
```

1.7 Stage 6: GA4GH v3.5 Stratification, GIAB v5.0Q/CMRG/Mosaic, HG008 v0.4 11-Caller Matrix, and Low-VAF Titration (`s6_*`)

Stage 6 comprises `4` cells (`s6_ga4gh_v35_stratification_matrix` , `s6_hg002_v50q_cmrg_mosaic_eval` , `s6_hg008_modern_ml_and_commercial_callers` —including the official NIST GIAB `Liss_lab` somatic callsets for `DRAGEN v4.2.4` , `DeepSomatic v1.6.0` , and `ClairS v0.4.1` , alongside `Lancet2` (`NYGC`), `Hartwig SAGE` , and `Ultima DeepVariant Somatic` —and `s6_low_vaf_titration_truth_curve`):

```
# Controlled in silico tumor-purity titration (20%, 10%, 5%, 2%, 1% purity at -f 0.002, s6_low_vaf_titration_truth_curve):
for PURITY in 20 10 5 2 1; do
  vardictrust -G GRCh38_GIABv3.fasta -f 0.002 -N "HG008_P${PURITY}|HG008-N" \
    -b "hg008_titration_p${PURITY}.bam|HG008-N.bam" -th 16 --fisher -z -c 1 -S 2 -E 3 -g 4 \
    --vcf hg008_slice.bed > "rust_titration_p${PURITY}.vcf"
done

# Next-gen GIAB v5.0Q T2T-Q100, CMRG v1.00, and Mosaic v1.10 evaluation (s6_hg002_v50q_cmrg_mosaic_eval):
rtg vcfeval -b HG002_GRCh38_v5.0q_T2T.vcf.gz -e HG002_GRCh38_v5.0q_T2T.bed \
  -c hg002_arm_a.vcf.gz -t GRCh38.sdf -o eval_v50q_arm_a
```

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2. Supplementary Note 1: Analytical Derivation of JVM (`VarDictJava`) and `vardictcpp` Heap Retention (`Arm A` vs `Arm B`)

In `VarDictJava` v1.8.3 (`CigarParser.java` and `VariationRealigner.java`), the pileup engine processes each BED region *[S, E]* of length $L = E - S + 1$ by instantiating two region-wide heap structures before emitting a single variant row:

1. A reference base map `Reference.referenceSequences` (`Map<Integer, Character>`) containing *O(L)* boxed `java.lang.Integer` (`16` bytes) and `java.lang.Character` (`16` bytes) objects plus `HashMap$Node` headers (`32` bytes), consuming ≈ *64–96* bytes per reference base (≈ *16–24* GiB for human `chr1` of length *L = 249,250,621* bp per worker thread).
2. A position-indexed variation map `Map<Integer, VariationMap<String, Variation>> nonInsertionVariants` and `insertionVariants`, where every covered genomic coordinate $p \in [S, E]$ allocates a `VariationMap` and one `Variation` object per observed allele (including the reference allele). At sequencing depth *D*, the active heap footprint per worker thread scales as:

$$\mathrm{RSS}_{\text{Java / C++}}(L, T) = \Theta\left(\Sigma_{t=1}^{T} L_t \cdot (c_{\text{ref}} + c_{\text{var}} \cdot \bar{a}(D))\right)$$

where L_t is the length of the region assigned to thread t and $\bar{a}(D)$ is the mean number of candidate alleles per locus. When $L_t \le 50$ kbp (Arm B), L_t is small enough that GC can reclaim objects between intervals (`34.14` GB on HG002 Arm B; `62.51` GB with timeout `rc=124` on HG008 Arm B). When $L_t \in [50, 249]$ Mbp (Arm A un-chunked chromosomes or `-R chr1:1-249250621`), live objects cannot be garbage-collected until the entire chromosome completes, driving `VarDictJava` to `106.47` GiB on `chr1` alone and `>213.7` GB (`rc=3` `OutOfMemoryError`) on multi-threaded full genomes (`Table 2`, `Table 5`). Similarly, `vardictcpp` retains per-region C++ `robin_hood` / `std::unordered_map` structures across each interval, consuming `10.58` GiB on `chr1`, `31.91` GB on HG002 Arm A, and `91.85` GB (`rc=124` timeout) on HG008 Arm A unless an external shell script pre-splits the BED file (`CPP_CHUNK=50,000`).

By contrast, `VarDictRust` leaves intervals ≤ *1,000,000* bp (`DEFAULT_UNTILED_SPAN_MAX`) untiled and partitions any region of length *L > 1,000,000* bp into fixed tiles of default width *W = 100* kbp (`DEFAULT_TILE_SIZE`) with halo *H = 10* kbp (`200` kbp in `-i` splice mode; `Figure 1b`), bounding memory strictly to:

$$\mathrm{RSS}_{\text{VarDictRust}}(L, T) = O(T \cdot (W + 2H)) \perp L$$

Empirically, increasing *L* by `250`× on chromosome 1 (`1` Mbp to `249` Mbp, `s4_span_chr1_3way_th8`..`th32`) changes `VarDictRust` RSS by only `1.18`×–`1.41`× (`1.38`–`3.69` GiB), enabling un-chunked full-genome calling in `4.93` GB (HG002) and `12.14` GB (HG008).

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3. Supplementary Note 2: Ten-Point Technical Audit of `vardictcpp` Divergences (`CPP-1` through `CPP-10`)

Across cells `s2_tierp_mode_parity_sweep`, `s5_h4_maintenance_security_matrix`, `s5_wgs_hg002_full_contigs_3way_th32`, and `s6_low_vaf_titration_truth_curve`, we cataloged ten reproducibility and capability gaps in `vardictcpp ae93fe39dc (Wolff 2026)`, matching the taxonomy in `Table S6` verbatim (`Table 8`, `Table S6`):

- **`CPP-1` (VCF Header & `INFO/END` Schema):** `vardictcpp` omits the `##INFO=<ID=END>` header definition and diverges from `var2vcf_valid.pl` / `var2vcf_paired.pl` VCF 4.3 schema formatting.
- **`CPP-2` (CRISPR (`-J` / `-j`) & Structural Variant Modes):** `vardictcpp` sets `--nosv` (`-U`) by default and omits CRISPR cleavage-site (`-J` / `-j`) and structural variant (`StructuralVariantsProcessor`) assembly (`1/13` modes failed in `s2_tierp_mode_parity_sweep`, `12/13` exact).
- **`CPP-3` (Amplicon (`-a`) Output Format):** In amplicon primer-clipping mode (`-a 10:0.95`, `s1_ceiling_amplicon`), `vardictcpp` emits raw tab-separated `.var` text instead of VCF (`0` VCF lines), breaking downstream `bcftools` and `rtg vcfeval` consumers.
- **`CPP-4` (Concatenated CLI Flag Parsing):** Passing bundled single-character flags (e.g., `-z -c 1` combined or legacy `VarDictJava` bundled options) causes `vardictcpp` to fail CLI parsing with exit code `120`.
- **`CPP-5` (50 kbp Chunk Boundary Deduplication):** Without an external BED-splitting wrapper (`--chunk 50,000` / `CPP_CHUNK=50,000`), `vardictcpp` consumes `10.58` GiB on `chr1`, `31.91` GB on HG002 Arm A, and `91.85` GB (`rc=124` timeout) on HG008 Arm A; when intervals are chunked, `vardictcpp` lacks boundary ownership deduplication, emitting `+309,569` excess duplicate/boundary-shifted calls on HG002 Arm A (`9,953,338` vs `9,643,769`) and `+394` on Arm B (`8,111,565` vs `8,111,171`).
- **`CPP-6` (Missing `##contig` Header Lines):** `vardictcpp` omits `##contig=<ID=...,length=...>` lines from VCF headers (`VarDictJava` issue `#320`), causing strict VCF validators and `rtg vcfeval` header checks to warn or fail.
- **`CPP-7` (GRCh37 / b37 Chromosome Naming):** `vardictcpp` hardcodes `chr`-prefix assumptions in internal chromosome handling, breaking or diverging on `b37` / `GRCh37` numeric contigs (`1`..`22`, `X`, `Y`, `MT`).
- **`CPP-8` (BAM/CRAM Reference & Decode):** `vardictcpp` diverges on CRAM reference decoding (`-G ref.fa`) and requires dynamic linkage against external `libhts`, `libcurl`, `liblzma`, `libbz2`, and `libdeflate`, whereas `VarDictRust` natively decodes BAM and CRAM in a `4.56` MB zero-dependency static `musl` binary (`s4_multimodal_and_cram_twin`).
- **`CPP-9` (MNP Phasing, Tie-Break & Somatic SV Drops):** `vardictcpp` diverges from `VarDictJava` on complex multinucleotide variant (MNP) phasing, realignment tie-breaking, and somatic variant retention—most visibly at `20%` tumor purity (`s6_low_vaf_titration_truth_curve`), where `vardictcpp` drops somatic indel F_1 to `0.2466` (recall `15.65%`) and SNV F_1 to `0.6914` (recall `57.07%`), whereas `VarDictRust` matches `VarDictJava` (`Indel` $F_1 = 0.3104$, recall `20.75%`; `SNV` $F_1 = 0.7512$, recall `65.59%`).
- **`CPP-10` (RNA-Seq Splice Mode (`-i` `.var` / `.splice.tsv`):** `vardictcpp` does not implement `-i` exon-junction splice-site accounting (`s3_rna_splice_parity_n3`, `51.8` GiB RNA-seq BAM, `12,668` records).

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4. Supplementary Note 3: Biostatistical Methodology, Exact Clopper-Pearson Intervals, and `--fix` Governance

4.1 Exact Clopper-Pearson Binomial Confidence Intervals

For all binary oracle-concordance and completion rates (k successes out of n independent evaluation cells/arms), we compute exact two-sided $1 - \alpha = 95\%$ Clopper-Pearson binomial confidence intervals $[p_{lo}, p_{hi}]$ via the inverse Beta distribution:

$$p_{lo} = \text{Beta}^{-1}(\alpha/2;\ k,\ n - k + 1), \qquad p_{hi} = \text{Beta}^{-1}(1 - \alpha/2;\ k + 1,\ n - k)$$

with $p_{lo} = 0$ when $k = 0$ and $p_{hi} = 1$ when $k = n$:

- Complete `60`-cell verification campaign (`60/60` exit `0`): `100%` (`95%` CI `94.04%–100.00%`)
- Stage 2 `13`-mode functional parity sweep (`13/13`): `100%` (`95%` CI `75.29%–100.00%`)
- Extended coverage Tier-2 modalities (`17/17`): `100%` (`95%` CI `80.49%–100.00%`)
- Multimodal Tier-2 slices (`18/18`): `100%` (`95%` CI `81.47%–100.00%`)
- Cross-platform chr20 runs where `VarDictJava` completed (`11/11`): `100%` (`95%` CI `71.51%–100.00%`)
- Cross-platform `-f 0` somatic noise-floor `STRICT-BYTES` sweep (`18/18`): `100%` (`95%` CI `81.47%–100.00%`)
- GA4GH `v3.5` `70`-cell stratification matrix (`70/70`): `100%` (`95%` CI `94.87%–100.00%`)
- Option C parity sweep across all parameter-sweep arms (`84/87`): `96.55%` (`95%` CI `90.25%–99.28%` , with `100%` [`84/84`] exact match across all standard ≤ *1,000,000* bp BED-interval sub-arms across all `10` BED-mode sweep cells, and `3` un-chunked > *1,000,000* bp single-interval sub-arms in `sweep_span_chr1_f0` [which achieves `100%` exact match across all `7,032` variants when run with `--tile-size 10000000` in `s4_span_chr1_f0`] and `sweep_R_chr1_full` [where untiled `249` Mbp `VarDictJava` pairs SVs across `36` – `81` Mbp centromeric distances and leaks `VD = 74 > DP` into `3` short deletions with impossible `740%` – `1850%` allele frequencies]; `Table S2`)
- NIST Mosaic SNV `v1.10` sensitivity (`84/85`): `98.82%` (`95%` CI `93.62%–99.97%`)

4.2 Non-Parametric Runtime and Memory Hypothesis Testing

Across the $N = 12$ Tier-P sample-by-thread cells (`s2_tierp_v2_*` , each measured across `n = 3` technical replicates), we report both the cell mean and full replicate range (`min–max` , `Table S3`) rather than normal-theory standard errors (which are unstable at $n = 3$). Comparing `VarDictRust` against `VarDictJava` across the $N = 12$ cells yields two-sided Mann-Whitney $U = 4.0$ ($p = 9.735e\text{-}05$) for wall-clock time and $U = 0.0$ ($p = 3.658e\text{-}05$) for peak RSS, and an exact paired sign-flip permutation test p-value of $p = 2^{-12} = 0.00024$.

4.3 Release Governance: `v1.0.x` Parity Versus `v1.1+` Opt-In `--fix` Flags

Under `ROADMAP.md` , all `v1.0.x` releases (`v1.0.6`) enforce strict oracle parity on valid inputs. True output-changing upstream bug fixes (e.g., `#92` `VD > DP` , `#62` over-filtering, `#292` `SHIFT3` , `#342` `<dup-N>` symbolic alleles) are gated behind explicit `--fix <id>` flags starting in `v1.1.0` , while parity-restoring fixes (`splice-tile-ownership` , `mate-window-ownership`) ship as verified defaults.

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5. Supplementary Tables (`Table S1` – `Table S6`) and Complete 60-Cell Verification Ledger

Supplementary Table S1 (Table S1). Complete 60-Directory GCP Study Inventory, Exit Codes & Paper Target Mapping

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
s1_ceiling_amplicon	stage1	Tier P	0	6	Fig 1, Fig 4, Table 1, Supp Table S1	MATCH (2 variants)	EGFR/MRE11a amplicon (-a) 3-way ceiling & exact parity
s1_ceiling_hg008	stage1	Tier G	0	6	Fig 1, Fig 4, Table 1, Supp Table S1	MATCH (18426 variants)	NIST GIAB HG008 somatic slice 3-way ceiling & exact parity
s1_gate7_ref_multiregion	stage1	Tier G	0	5	Fig 1, Fig 2, Table 1, Supp Table S1	MATCH (2572 variants)	HG002 chr20 multi-region reference boundary parity gate
s1_h1a_simple_tiles	stage1	Tier E	0	5	Fig 1, Fig 2, Table 1, Supp Table S1	MATCH (5090 variants); MATCH (5090 variants); MATCH (5090 variants); MATCH (5090 variants)	Tile-size invariance (0.5, 1, 2, 10 Mbp) on SEQC2 HCC1395

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
s2_fig1_fig3_field_callers	stage2	Tier P	0	73	Fig 1, Fig 2, Table 2, Supp Table S1	PASS_16_FIELD_CALLER_RUNS	Wolff 2026 Fig 1 & Fig 3 field callers (FreeBayes, LoFreq, Mutect2, octopus, Strelka2) at 1c/8c
s2_fig2_hg002_chr20_truth	stage2	Tier G	0	20	Fig 1, Fig 3, Table 2, Table 3, Supp Table S1	MATCH (8233 variants); MATCH_TRUTH_7_CALLERS (VarDictRust==VarDictJava SNV_F1=0.9858,INDEL_F1=0.8956)	Wolff 2026 Fig 2 / Daniels 2024 HG002 chr20 exome 7-caller truth accuracy
s2_table3_R_chr1_java_200g_n3	stage2	Tier P	0	6	Fig 1, Fig 4, Table 2, Table 4, Supp Table S1	JAVA_XMX4G_OOM_RC3;JAVA_XMX200G_PASS_N3	Wolff 2026 Table 3 -R chr1:1-249250621 Java -Xmx4g OOM & -Xmx200g completion (n=3)
s2_table3_R_chr1_rust_and_cpp	stage2	Tier P	0	10	Fig 1, Fig 4, Table 2, Table 4, Supp Table S1	PASS_R_CHR1_RUST_AND_CPP_N3	Wolff 2026 Table 3 -R chr1:1-249250621 VarDictRust vs vardictcpp (n=3)
s2_tierp_mode_parity_sweep	stage2	Tier P	0	14	Fig 1, Fig 2, Fig 8, Table 1, Table 2, Table 8, Supp Table S1, Supp Table S6	MATCH=13/13_MODES	13-mode 3-way functional parity sweep (Simple, Somatic, Amplicon, Splice, SV, CRAM, Fisher, -f 0)
s2_tierp_v2_SRR15006375_th1	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (4056 rows); MATCH (3966 variants)	Tier-P AML panel SRR15006375 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006375_th8	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (4056 rows); MATCH (3966 variants)	Tier-P AML panel SRR15006375 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR15006376_th1	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (12739 rows); MATCH (12655 variants)	Tier-P AML panel SRR15006376 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006376_th8	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (12739 rows); MATCH (12655 variants)	Tier-P AML panel SRR15006376 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR15006386_th1	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (4014 rows); MATCH (3925 variants)	Tier-P AML panel SRR15006386 3-way parity & timing at 1 thread (n=3)

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
s2_tierp_v2_SRR15006386_th8	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (4014 rows); MATCH (3925 variants)	Tier-P AML panel SRR15006386 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR15006540_th1	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (16031 rows); MATCH (15912 variants)	Tier-P AML panel SRR15006540 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR15006540_th8	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (16031 rows); MATCH (15912 variants)	Tier-P AML panel SRR15006540 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_SRR8657348_th1	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (54448 rows); MATCH (53224 variants)	Tier-P AML panel SRR8657348 (773,667 intervals) 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_SRR8657348_th8	stage2	Tier P	0	19	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (54448 rows); MATCH (53224 variants)	Tier-P AML panel SRR8657348 (773,667 intervals) 3-way parity & timing at 8 threads (n=3)
s2_tierp_v2_somatic_76_86_th1	stage2	Tier P	0	17	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (14126 rows); MATCH (14000 variants)	Tier-P somatic pair SRR15006376/SRR15006386 3-way parity & timing at 1 thread (n=3)
s2_tierp_v2_somatic_76_86_th8	stage2	Tier P	0	17	Fig 1, Fig 2, Table 2, Supp Table S1, Supp Table S3	MATCH (14126 rows); MATCH (14000 variants)	Tier-P somatic pair SRR15006376/SRR15006386 3-way parity & timing at 8 threads (n=3)
s3_rna_splice_parity_n3	stage3	Tier G (RNA)	0	10	Fig 1, Fig 4, Table 1, Table 4, Supp Table S1	MATCH (12668 rows)	Full-sample 51.8 GiB HG002 GM24385 RNA-seq (-i splice mode) parity & timing (n=3)
s4_jobs_per_node_2x4c_n3	stage4	Tier E	0	7	Fig 1, Fig 4, Table 4, Supp Table S1	PASS_2X4C_NODE_PACKING_N3	Concurrent 2x4-core node-packing throughput & RSS on SEQC2 HCC1395 exome (n=3)
s4_multimodal_and_cram_twin	stage4	Tier G	0	3	Fig 1, Fig 2, Table 1, Supp Table S1	STRICT-BYTES: 0 textual difference(s) within eps	Lossless BAM-vs-CRAM twin zero-diff check & version verification
s4_perf_budget_4targets_n3	stage4	Tier P/E/G	0	54	Fig 1, Fig 4, Table 4, Supp Table S1	PASS_4TARGETS_N3	3-way performance budget across amplicon, s3_simple, s4_somatic, hg008_somatic (n=3)

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
s4_scale_exome_3way_th1	stage4	Tier E	0	11	Fig 1, Fig 4, Fig 8, Table 4, Supp Table S1, Supp Table S3	MATCH (339698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 1 thread (n=3)
s4_scale_exome_3way_th2	stage4	Tier E	0	12	Fig 1, Fig 4, Fig 8, Table 4, Supp Table S1, Supp Table S3	MATCH (339698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 2 threads (n=3)
s4_scale_exome_3way_th4	stage4	Tier E	0	12	Fig 1, Fig 4, Fig 8, Table 4, Supp Table S1, Supp Table S3	MATCH (339698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 4 threads (n=3)
s4_scale_exome_3way_th8	stage4	Tier E	0	12	Fig 1, Fig 4, Fig 8, Table 4, Supp Table S1, Supp Table S3	MATCH (339698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 8 threads (n=3)
s4_scale_exome_3way_th16	stage4	Tier E	0	12	Fig 1, Fig 4, Fig 8, Table 4, Supp Table S1, Supp Table S3	MATCH (339698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 16 threads (n=3)
s4_scale_exome_3way_th32	stage4	Tier E	0	12	Fig 1, Fig 4, Fig 8, Table 4, Supp Table S1, Supp Table S3	MATCH (339698 variants)	SEQC2 HCC1395 whole-exome 3-way scaling at 32 threads (n=3)
s4_span_chr1_3way_th8	stage4	Tier E	0	37	Fig 1, Fig 4, Table 4, Supp Table S1, Supp Table S3	PASS_SPAN_1M_TO_249M_N3	Chromosome 1 span-to-RSS scaling (1, 10, 100, 249 Mbp) at 8 threads (n=3)
s4_span_chr1_3way_th16	stage4	Tier E	0	37	Fig 1, Fig 4, Table 4, Supp Table S1, Supp Table S3	PASS_SPAN_1M_TO_249M_N3	Chromosome 1 span-to-RSS scaling (1, 10, 100, 249 Mbp) at 16 threads (n=3)
s4_span_chr1_3way_th32	stage4	Tier E	0	37	Fig 1, Fig 4, Table 4, Supp Table S1, Supp Table S3	PASS_SPAN_1M_TO_249M_N3	Chromosome 1 span-to-RSS scaling (1, 10, 100, 249 Mbp) at 32 threads (n=3)

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
s4_span_chr1_f0	stage4	Tier E	0	9	Fig 1, Fig 4, Table 1, Table 4, Supp Table S1	MATCH (7032 variants)	Chromosome 1 10 Mbp all-positions (-f 0) parity & memory stress (n=3)
s4_tier3_truth_and_qual_sweep	stage4	Tier E/G	0	24	Fig 1, Fig 3, Fig 7, Table 3, Table 7, Supp Table S1, Supp Table S4	MATCH_TRUTH_3_COHORTS_AND_QUAL_ROC	rtg vcfeval truth concordance & QUAL ROC sweep (0..200) on HG002, SEQC2, HG008
evidence_certified	stage4	Tier P/E/G	0	16	Fig 1, Fig 2, Fig 3, Table 1, Table 3, Supp Table S1	PASS_4_GCP_SUITES	GCP-certified evidence bundle (17-modality extended coverage, 18-slice multimodal, HG005, NA24385-b37)
sweep_4targets_full	sweep	Tier P/E/G	0	15	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (4/4 sub-arms exact match)	Option C .var & .vcf parity sweep across amplicon, hg002, seqc2, hg008
sweep_R_chr1_full	sweep	Tier P	0	6	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH=0/1 sub-arms (1 unchunked >1 Mbp auto-tiled vs untiled Java diffs; see Table S2)	Option C .var parity sweep on -R chr1:1-249250621 (249 Mbp single region)
sweep_hg002_chr20_exome	sweep	Tier G	0	9	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (2/2 sub-arms exact match)	Option C .var & .vcf parity sweep on HG002 chr20 exome (Daniels 2024)
sweep_multimodal_and_cram_twin	sweep	Tier G	0	11	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (4/4 sub-arms exact match)	Option C multimodal slices, extended coverage, and HG008 CRAM twin timing & parity
sweep_rna_hg002_52gb	sweep	Tier G (RNA)	0	10	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (2/2 sub-arms exact match)	Option C .var & .splice.tsv parity sweep on 51.8 GiB HG002 RNA-seq
sweep_span_chr1_f0	sweep	Tier E	0	9	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH=0/2 sub-arms (2 unchunked >1 Mbp auto-tiled vs untiled Java diffs; see Table S2)	Option C .var & .vcf parity sweep on chr1 10 Mbp at -f 0
sweep_tierp_13modes	sweep	Tier P	0	49	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (12/12 sub-arms exact match)	Option C .var parity sweep across 11 Tier-P operational modes

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
sweep_tierp_SRR15006540	sweep	Tier P	0	9	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (2/2 sub-arms exact match)	Option C .var & .vcf parity sweep on Tier-P SRR15006540
sweep_tierp_SRR8657348	sweep	Tier P	0	9	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (2/2 sub-arms exact match)	Option C .var & .vcf parity sweep on Tier-P SRR8657348
sweep_tierp_somatic_76_86	sweep	Tier P	0	9	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (2/2 sub-arms exact match)	Option C .var & .vcf parity sweep on Tier-P somatic pair SRR15006376/SRR15006386
sweep_ultradeep_mrd_stress	sweep	Tier P/G	0	107	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (30/30 sub-arms exact match)	Option C ultra-deep MRD stress test (-f 0.0005 and -f 0 with --fisher up to 45,000x)
sweep_wgs_mrd_matrix	sweep	Tier E/G	0	86	Fig 1, Fig 2, Table 1, Supp Table S1, Supp Table S2	MATCH (24/24 sub-arms exact match)	Option C WGS & exome MRD matrix (-f 0.001 and -f 0 across HG002, HG005 300x, SEQC2, HG008)
s5_cross_platform_short_read_chr20_panel	stage5	Tier G (10-Platform)	0	420	Fig 1, Fig 6, Table 6, Supp Table S1, Supp Table S4	STRICT-BYTES: 0 textual difference(s) within eps; STRICT-BYTES: 0 textual difference(s) within eps; MATCH_10_PLATFORMS_AND_18_18_F0_NOISE_FLOOR	10-platform sequencing chemistry chr20 panel (Part A -f 0.01/-f 0 std/UG-HCR; Part B 18-arm noise floor; Ultima CRAM twin)
s5_grch38_field_callers_truth	stage5	Tier G	0	78	Fig 1, Fig 3, Table 3, Supp Table S1	PASS_19_CALLER_COHORT_EVALS (VarDictRust==VarDictJava)	Native GRCh38 germline (HG002 chr20) & somatic (HG008 slice, SEQC2) field-caller truth evaluation
s5_h4_maintenance_security_matrix	stage5	All Tiers	0	13	Fig 1, Fig 8, Table 8, Supp Table S1, Supp Table S6	PASS_H4_STATIC_MUSL_AND_CPP10_CATALOG	H4 supply-chain security, static musl binary, CLI flag parity, and CPP-1..10 divergence matrix
s5_wgs_hg002_full_contigs_3way_th32	stage5	Tier G (WGS)	0	14	Fig 1, Fig 5, Table 5, Supp Table S1	OOM-vs-completed (java_rc=3, rust_vars=9643769)	Full-genome HG002 (130.8 GB 2x250 bp) Arm A (22 un-chunked chromosomes) 3-way scaling
s5_wgs_hg002_full_giab_3way_th32	stage5	Tier G (WGS)	0	27	Fig 1, Fig 5, Table 1, Table 5, Supp Table S1	MATCH (8111171 variants)	Full-genome HG002 (130.8 GB 2x250 bp) Arm B (≤50 kbp GIAB callable intervals) 3-way parity & truth
s5_wgs_hg008_full_contigs_3way_th32	stage5	Tier G (WGS)	0	12	Fig 1, Fig 5, Table 5, Supp Table S1	OOM-or-timeout-vs-completed (java_rc=3, rust_vars=13067287)	Full-genome HG008 (329.0 GB 364x T/N) Arm A (22 un-chunked chromosomes) 3-way scaling

Cell ID	Stage	Tier	Exit Code	Files	Paper Targets	Parity / Execution Verdict	Description
s5_wgs_hg008_full_giab_3way_th32	stage5	Tier G (WGS)	0	20	Fig 1, Fig 5, Table 5, Supp Table S1	OOM-or-timeout-vs-completed (java_rc=124, rust_vars=10195118)	Full-genome HG008 (329.0 GB 364x T/N) Arm B (≤50 kbp GIAB v0.4 intervals) 3-way scaling & v0.3/v0.4 truth
s6_ga4gh_v35_stratification_matrix	stage6	Tier G (GA4GH v3.5)	0	2	Fig 1, Fig 6, Table 6, Supp Table S1, Supp Table S5	MATCH_70_STRATIFICATION_CELLS	70-cell GA4GH v3.5 genomic stratification matrix (7 strata x 10 sequencing platforms)
s6_hg002_v50q_cmrg_mosaic_eval	stage6	Tier G (v5.0Q/CMRG/Mosaic)	0	2	Fig 1, Fig 7, Table 7, Supp Table S1	MATCH_V50Q_CMRG_MOSAIC_12_EVALS	HG002 GIAB v5.0Q T2T-Q100, CMRG v1.00, and Mosaic SNV v1.10 evaluation
s6_hg008_modern_ml_and_commercial_callers	stage6	Tier G (HG008 v0.4)	0	2	Fig 1, Fig 7, Table 7, Supp Table S1	PASS_27_CALLER_BENCHMARKS_V04	HG008 GIAB v0.4 benchmark across VarDictRust, 7 NIST Liss_lab callers (DRAGEN, DeepSomatic, ClairS, Lancet2, SAGE, Ultima DV), Mutect2, Strelka2, LoFreq
s6_low_vaf_titration_truth_curve	stage6	Tier E (Titration)	0	42	Fig 1, Fig 3, Table 3, Supp Table S1	MATCH (17161 variants); MATCH (16350 variants); MATCH (17593 variants); MATCH (16512 variants); MATCH (16857 variants); MATCH_30_TITRATION_EVALS (VarDictRust==VarDictJava)	HG008 somatic tumor-purity titration (20%→1% purity / 10%→0.5% VAF) at -f 0.002 across 6 callers

Every directory in gs://vardict-paper-trial-results/cells-v106/ completed with exit_code=0 and maps to at least one main figure and table.

Supplementary Table S2 (Table S2). Baseline Equivalence Cells and Complete 87-Arm Full-Core .var & .vcf Parity Sweep

Cell ID	Sub-Arm / Equivalence Scope	VarDictJava Rows/Vars	VarDictRust Rows/Vars	FP vs. Java	FN vs. Java	Verdict
s1_ceiling_amplicon	EGFR/MRE11a amplicon (-a) ceiling & parity	MATCH (2 variants)	MATCH (2 variants)	0	0	MATCH
s1_ceiling_hg008	NIST GIAB HG008 somatic slice ceiling & parity	MATCH (18426 variants)	MATCH (18426 variants)	0	0	MATCH
s1_gate7_ref_multiregion	HG002 Chr20 multi-region reference offset verification	MATCH (2572 variants)	MATCH (2572 variants)	0	0	MATCH
s1_h1a_simple_tiles	SEQC2 HCC1395 tile-size invariance (0.5, 1, 2, 10 Mbp tiles)	5,090	5,090	0	0	MATCH across all 4 tile sizes
s4_multimodal_and_cram_twin	HG008 lossless BAM-vs-CRAM twin & multimodal slices	1,129	1,129	0	0	STRICT-BYTES: 0 textual difference(s) within eps; MATCH (40 variants)
sweep_4targets_full	amplicon_full.vcf	2	2	0	0	MATCH
sweep_4targets_full	hg002_deep_wgs.vcf	227,562	227,562	0	0	MATCH
sweep_4targets_full	hg008_somatic_wgs.vcf	18,426	18,426	0	0	MATCH
sweep_4targets_full	seqc2_somatic_exome_full.vcf	623,444	623,444	0	0	MATCH
sweep_R_chr1_full	R_chr1_249Mbp.var	4,705	4,692	5	18	DIFF_CAPTURED
sweep_hg002_chr20_exome	hg002_chr20_exome.var	8,330	8,330	0	0	MATCH
sweep_hg002_chr20_exome	hg002_chr20_exome.vcf	8,233	8,233	0	0	MATCH
sweep_multimodal_and_cram_twin	extended_coverage_17modes	17	17	0	0	MATCH
sweep_multimodal_and_cram_twin	multimodal_18slices	18	18	0	0	MATCH
sweep_multimodal_and_cram_twin	cram_twin_g4_hg008	1	1	0	0	MATCH
sweep_multimodal_and_cram_twin	e2_cram_vs_bam_twin.vcf	40	40	0	0	MATCH
sweep_rna_hg002_52gb	rna_splice_table	12,668	12,668	0	0	MATCH
sweep_rna_hg002_52gb	rna_cdna_variants.var	1,230	1,230	0	0	MATCH
sweep_span_chr1_f0	chr1_10m_f0.var	7,077	7,079	3	1	DIFF_CAPTURED
sweep_span_chr1_f0	chr1_10m_f0.vcf	7,032	7,034	3	1	DIFF_CAPTURED
sweep_tierp_13modes	simple_f0.01.var	228	228	0	0	MATCH
sweep_tierp_13modes	simple_f0_mrd.var	248	248	0	0	MATCH
sweep_tierp_13modes	simple_fisher.var	650	650	0	0	MATCH
sweep_tierp_13modes	targeted_panel_chr17.var	83	83	0	0	MATCH
sweep_tierp_13modes	sv_deletion_regions.var	3	3	0	0	MATCH

Cell ID	Sub-Arm / Equivalence Scope	VarDictJava Rows/Vars	VarDictRust Rows/Vars	FP vs. Java	FN vs. Java	Verdict
sweep_tierp_13modes	low_coverage.var	146	146	0	0	MATCH
sweep_tierp_13modes	amplicon_c17amp.var	145	145	0	0	MATCH
sweep_tierp_13modes	amplicon_fisher.var	145	145	0	0	MATCH
sweep_tierp_13modes	somatic_f0.01_big.var	664	664	0	0	MATCH
sweep_tierp_13modes	somatic_f0_mrd_c17.var	1,396	1,396	0	0	MATCH
sweep_tierp_13modes	somatic_fisher_c17.var	527	527	0	0	MATCH
sweep_tierp_13modes	P5_cram_vs_bam_twin.var	228	228	0	0	MATCH
sweep_tierp_SRR15006540	SRR15006540.var	16,031	16,031	0	0	MATCH
sweep_tierp_SRR15006540	SRR15006540.vcf	15,912	15,912	0	0	MATCH
sweep_tierp_SRR8657348	SRR8657348.var	54,448	54,448	0	0	MATCH
sweep_tierp_SRR8657348	SRR8657348.vcf	53,224	53,224	0	0	MATCH
sweep_tierp_somatic_76_86	somatic_76_86.var	14,126	14,126	0	0	MATCH
sweep_tierp_somatic_76_86	somatic_76_86.vcf	14,000	14,000	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006375_f0.0005_fisher.var	2,780	2,780	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006375_f0.0005_fisher.vcf	2,645	2,645	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006375_f0_fisher.var	6,182	6,182	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006375_f0_fisher.vcf	5,685	5,685	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006540_f0.0005_fisher.var	5,046	5,046	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006540_f0.0005_fisher.vcf	4,711	4,711	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006540_f0_fisher.var	7,044	7,044	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006540_f0_fisher.vcf	6,313	6,313	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006386_f0.0005_fisher.var	3,028	3,028	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006386_f0.0005_fisher.vcf	2,888	2,888	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006386_f0_fisher.var	6,153	6,153	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006386_f0_fisher.vcf	5,608	5,608	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006376_f0.0005_fisher.var	2,844	2,844	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006376_f0.0005_fisher.vcf	2,732	2,732	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006376_f0_fisher.var	3,841	3,841	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_SRR15006376_f0_fisher.vcf	3,630	3,630	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_76_86_29kx_f0.0005_fisher.var	4,045	4,045	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_76_86_29kx_f0.0005_fisher.vcf	3,855	3,855	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_76_86_29kx_f0_fisher.var	6,890	6,890	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_76_86_29kx_f0_fisher.vcf	6,309	6,309	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_75_86_45kx_f0.0005_fisher.var	4,481	4,481	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_75_86_45kx_f0.0005_fisher.vcf	4,262	4,262	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_75_86_45kx_f0_fisher.var	7,474	7,474	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_som_75_86_45kx_f0_fisher.vcf	6,828	6,828	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_hg008_15kx_f0.0005_fisher.var	19	19	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_hg008_15kx_f0.0005_fisher.vcf	16	16	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_hg002_7kx_f0.0005_fisher.var	1,008	1,008	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_hg002_7kx_f0.0005_fisher.vcf	640	640	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_rna_173kx_f0.001_fisher.var	139	139	0	0	MATCH
sweep_ultradeep_mrd_stress	ud_rna_173kx_f0.001_fisher.vcf	120	120	0	0	MATCH
sweep_wgs_mrd_matrix	hg002_wgs_mrd_f0.001.var	15,083	15,083	0	0	MATCH
sweep_wgs_mrd_matrix	hg002_wgs_mrd_f0.001.vcf	14,769	14,769	0	0	MATCH
sweep_wgs_mrd_matrix	hg002_wgs_mrd_f0.var	15,083	15,083	0	0	MATCH
sweep_wgs_mrd_matrix	hg002_wgs_mrd_f0.vcf	14,769	14,769	0	0	MATCH
sweep_wgs_mrd_matrix	hg008_somatic_wgs_mrd_f0.001.var	21,379	21,379	0	0	MATCH
sweep_wgs_mrd_matrix	hg008_somatic_wgs_mrd_f0.001.vcf	20,293	20,293	0	0	MATCH
sweep_wgs_mrd_matrix	hg008_somatic_wgs_mrd_f0.001_fisher.var	21,379	21,379	0	0	MATCH
sweep_wgs_mrd_matrix	hg008_somatic_wgs_mrd_f0.001_fisher.vcf	20,293	20,293	0	0	MATCH
sweep_wgs_mrd_matrix	hg008_somatic_wgs_mrd_f0.var	21,379	21,379	0	0	MATCH
sweep_wgs_mrd_matrix	hg008_somatic_wgs_mrd_f0.vcf	20,293	20,293	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr3_f0.001.var	339	339	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr3_f0.001.vcf	325	325	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr3_f0.var	339	339	0	0	MATCH

Cell ID	Sub-Arm / Equivalence Scope	VarDictJava Rows/Vars	VarDictRust Rows/Vars	FP vs. Java	FN vs. Java	Verdict
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr3_f0.vcf	325	325	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr6_f0.001.var	305	305	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr6_f0.001.vcf	299	299	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr21_f0.001.var	286	286	0	0	MATCH
sweep_wgs_mrd_matrix	hg005_300x_wgs_chr21_f0.001.vcf	284	284	0	0	MATCH
sweep_wgs_mrd_matrix	seqc2_somatic_mrd_f0.001.var	5,261	5,261	0	0	MATCH
sweep_wgs_mrd_matrix	seqc2_somatic_mrd_f0.001.vcf	5,206	5,206	0	0	MATCH
sweep_wgs_mrd_matrix	seqc2_somatic_mrd_f0.001_fisher.var	5,261	5,261	0	0	MATCH
sweep_wgs_mrd_matrix	seqc2_somatic_mrd_f0.001_fisher.vcf	5,206	5,206	0	0	MATCH
sweep_wgs_mrd_matrix	seqc2_somatic_mrd_f0.var	5,261	5,261	0	0	MATCH
sweep_wgs_mrd_matrix	seqc2_somatic_mrd_f0.vcf	5,206	5,206	0	0	MATCH

All 84/84 standard BED-interval sub-arms (<=1 Mbp) achieve 100% exact parity; the 3 DIFF_CAPTURED sub-arms occur exclusively on the two unchunked >1 Mbp single-region runs (chr1 10 Mbp auto-tiled at 100 kbp, which achieves 100% match with --tile-size 10000000 in s4_span_chr1_f0, and -R chr1 249 Mbp where untiled VarDictJava pairs SVs across 36-81 Mbp and leaks VD=74 > DP into 3 short deletions).

Supplementary Table S3 (Table S3). Complete Technical Replicate (n=3) and Full-WGS Wall-Clock and Peak RSS Ledger

Cell ID	Workload	Threads	Engine	Replicate Wall Times (s)	Wall Mean (Min-Max) (s)	Replicate Peak RSS (KB)	Peak RSS Mean (MB / GiB)
s2_tierp_v2_SRR15006375_th1	AML Panel SRR15006375	1	VarDictRust	121.14, 122.26, 122.38	121.93 (121.14-122.38)	215108, 215236, 215736	210.3 MB (0.205 GiB)
s2_tierp_v2_SRR15006375_th1	AML Panel SRR15006375	1	VarDictJava	404.89, 418.94, 400.74	408.19 (400.74-418.94)	2590208, 2591732, 2612224	2537.2 MB (2.478 GiB)
s2_tierp_v2_SRR15006375_th1	AML Panel SRR15006375	1	vardictcpp	109.90, 109.88, 109.19	109.66 (109.19-109.90)	65292, 65280, 65312	63.8 MB (0.062 GiB)
s2_tierp_v2_SRR15006375_th8	AML Panel SRR15006375	8	VarDictRust	49.39, 49.59, 49.44	49.47 (49.39-49.59)	278052, 274168, 277424	270.1 MB (0.264 GiB)
s2_tierp_v2_SRR15006375_th8	AML Panel SRR15006375	8	VarDictJava	229.14, 235.27, 233.69	232.70 (229.14-235.27)	4665344, 4511724, 4371936	4410.5 MB (4.307 GiB)
s2_tierp_v2_SRR15006375_th8	AML Panel SRR15006375	8	vardictcpp	15.42, 15.32, 15.30	15.35 (15.30-15.42)	342664, 334764, 331092	328.3 MB (0.321 GiB)
s2_tierp_v2_SRR15006376_th1	AML Panel SRR15006376	1	VarDictRust	80.84, 80.32, 80.08	80.41 (80.08-80.84)	231180, 231156, 230976	225.7 MB (0.220 GiB)
s2_tierp_v2_SRR15006376_th1	AML Panel SRR15006376	1	VarDictJava	326.10, 340.26, 336.12	334.16 (326.10-340.26)	4428032, 4031016, 4579804	4244.4 MB (4.145 GiB)
s2_tierp_v2_SRR15006376_th1	AML Panel SRR15006376	1	vardictcpp	83.78, 84.32, 83.03	83.71 (83.03-84.32)	75804, 75808, 75804	74.0 MB (0.072 GiB)
s2_tierp_v2_SRR15006376_th8	AML Panel SRR15006376	8	VarDictRust	48.86, 48.70, 48.65	48.74 (48.65-48.86)	287200, 288108, 287872	281.0 MB (0.274 GiB)
s2_tierp_v2_SRR15006376_th8	AML Panel SRR15006376	8	VarDictJava	174.29, 150.76, 168.04	164.36 (150.76-174.29)	7744512, 8044032, 7073792	7442.2 MB (7.268 GiB)
s2_tierp_v2_SRR15006376_th8	AML Panel SRR15006376	8	vardictcpp	13.09, 12.50, 12.96	12.85 (12.50-13.09)	351772, 364060, 371688	354.0 MB (0.346 GiB)
s2_tierp_v2_SRR15006386_th1	AML Panel SRR15006386	1	VarDictRust	114.04, 113.35, 113.17	113.52 (113.17-114.04)	191748, 191976, 192300	187.5 MB (0.183 GiB)
s2_tierp_v2_SRR15006386_th1	AML Panel SRR15006386	1	VarDictJava	334.16, 338.80, 331.87	334.94 (331.87-338.80)	2596316, 2423020, 2425344	2423.4 MB (2.367 GiB)
s2_tierp_v2_SRR15006386_th1	AML Panel SRR15006386	1	vardictcpp	98.37, 98.14, 98.09	98.20 (98.09-98.37)	57516, 57516, 57520	56.2 MB (0.055 GiB)
s2_tierp_v2_SRR15006386_th8	AML Panel SRR15006386	8	VarDictRust	43.81, 43.88, 43.64	43.78 (43.64-43.88)	312272, 312860, 313136	305.4 MB (0.298 GiB)
s2_tierp_v2_SRR15006386_th8	AML Panel SRR15006386	8	VarDictJava	178.99, 182.17, 180.70	180.62 (178.99-182.17)	4535524, 4536296, 4551680	4434.7 MB (4.331 GiB)
s2_tierp_v2_SRR15006386_th8	AML Panel SRR15006386	8	vardictcpp	16.69, 15.01, 14.95	15.55 (14.95-16.69)	300084, 309872, 307332	298.6 MB (0.292 GiB)
s2_tierp_v2_SRR15006540_th1	AML Panel SRR15006540	1	VarDictRust	125.49, 125.61, 126.91	126.00 (125.49-126.91)	199412, 199396, 199072	194.6 MB (0.190 GiB)
s2_tierp_v2_SRR15006540_th1	AML Panel SRR15006540	1	VarDictJava	406.72, 412.42, 402.18	407.11 (402.18-412.42)	4630248, 4671184, 4745216	4572.5 MB (4.465 GiB)
s2_tierp_v2_SRR15006540_th1	AML Panel SRR15006540	1	vardictcpp	111.55, 111.32, 111.66	111.51 (111.32-111.66)	78084, 78088, 78084	76.3 MB (0.074 GiB)
s2_tierp_v2_SRR15006540_th8	AML Panel SRR15006540	8	VarDictRust	77.89, 77.99, 78.24	78.04 (77.89-78.24)	233572, 227592, 229364	224.8 MB (0.220 GiB)
s2_tierp_v2_SRR15006540_th8	AML Panel SRR15006540	8	VarDictJava	179.54, 186.40, 187.21	184.38 (179.54-187.21)	3863552, 4506172, 4187860	4087.8 MB (3.992 GiB)

Cell ID	Workload	Threads	Engine	Replicate Wall Times (s)	Wall Mean (Min-Max) (s)	Replicate Peak RSS (KB)	Peak RSS Mean (MB / GiB)
s2_tierp_v2_SRR15006540_th8	AML Panel SRR15006540	8	vardictcpp	16.77, 16.31, 17.00	16.69 (16.31-17.00)	342792, 350980, 347656	339.0 MB (0.331 GiB)
s2_tierp_v2_SRR8657348_th1	AML Panel SRR8657348	1	VarDictRust	155.05, 155.12, 154.77	154.98 (154.77-155.12)	486840, 486476, 483624	474.3 MB (0.463 GiB)
s2_tierp_v2_SRR8657348_th1	AML Panel SRR8657348	1	VarDictJava	862.51, 857.96, 863.07	861.18 (857.96-863.07)	2079488, 2078976, 2073724	2028.7 MB (1.981 GiB)
s2_tierp_v2_SRR8657348_th1	AML Panel SRR8657348	1	vardictcpp	175.43, 173.98, 174.02	174.48 (173.98-175.43)	146848, 146932, 146936	143.5 MB (0.140 GiB)
s2_tierp_v2_SRR8657348_th8	AML Panel SRR8657348	8	VarDictRust	39.76, 39.62, 39.50	39.63 (39.50-39.76)	492940, 485740, 536324	493.2 MB (0.482 GiB)
s2_tierp_v2_SRR8657348_th8	AML Panel SRR8657348	8	VarDictJava	212.70, 217.72, 217.12	215.85 (212.70-217.72)	3713444, 3823360, 3822592	3697.7 MB (3.611 GiB)
s2_tierp_v2_SRR8657348_th8	AML Panel SRR8657348	8	vardictcpp	24.71, 24.68, 24.64	24.68 (24.64-24.71)	255472, 259568, 258544	251.8 MB (0.246 GiB)
s2_tierp_v2_somatic_76_86_th1	AML Panel somatic_76_86	1	VarDictRust	230.96, 230.70, 229.92	230.53 (229.92-230.96)	255464, 252304, 253316	247.8 MB (0.242 GiB)
s2_tierp_v2_somatic_76_86_th1	AML Panel somatic_76_86	1	VarDictJava	720.47, 728.53, 729.27	726.09 (720.47-729.27)	4415396, 4396032, 4420820	4307.4 MB (4.206 GiB)
s2_tierp_v2_somatic_76_86_th1	AML Panel somatic_76_86	1	vardictcpp	215.50, 215.04, 214.94	215.16 (214.94-215.50)	148252, 148252, 148504	144.9 MB (0.141 GiB)
s2_tierp_v2_somatic_76_86_th8	AML Panel somatic_76_86	8	VarDictRust	134.96, 133.83, 137.51	135.43 (133.83-137.51)	383724, 396596, 393488	382.1 MB (0.373 GiB)
s2_tierp_v2_somatic_76_86_th8	AML Panel somatic_76_86	8	VarDictJava	373.79, 371.55, 374.68	373.34 (371.55-374.68)	5850600, 5814784, 5783296	5679.9 MB (5.547 GiB)
s2_tierp_v2_somatic_76_86_th8	AML Panel somatic_76_86	8	vardictcpp	31.60, 31.64, 31.75	31.66 (31.60-31.75)	824788, 803560, 786332	786.0 MB (0.768 GiB)
s2_table3_R_chr1_java_200g_n3	SRR8657348 Unchunked - R chr1 (249 Mbp)	8	VarDictJava (-Xmx200g)	509.65, 492.93, 502.90	501.83 (492.93-509.65)	112714496, 111591168, 110633216	109029.6 MB (106.474 GiB)
s2_table3_R_chr1_rust_and_cpp	SRR8657348 Unchunked - R chr1 (249 Mbp)	8	VarDictRust	9.80, 9.81, 9.64	9.75 (9.64-9.81)	320192, 319576, 339012	318.6 MB (0.311 GiB)
s2_table3_R_chr1_rust_and_cpp	SRR8657348 Unchunked - R chr1 (249 Mbp)	8	vardictcpp	106.58, 105.07, 106.03	105.89 (105.07-106.58)	11098040, 11098040, 11098040	10837.9 MB (10.584 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	VarDictRust	135.43	135.43	391269	382.1 MB (0.373 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	VarDictJava	373.34	373.34	5816227	5679.9 MB (5.547 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	vardictcpp	31.66	31.66	804893	786.0 MB (0.768 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	Strelka2	199.96	199.96	1184452	1156.7 MB (1.130 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	LoFreq	521.82	521.82	2146108	2095.8 MB (2.047 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	octopus	450.87	450.87	2521004	2461.9 MB (2.404 GiB)
s2_fig1_fig3_field_callers	Somatic SRR15006376/86 Field Callers	8	Mutect2	969.01	969.01	2355632	2300.4 MB (2.247 GiB)
s3_rna_splice_parity_n3	51.8 GiB HG002 RNA-seq (-i splice)	8	VarDictRust	5.01, 4.90, 4.90	4.94 (4.90-5.01)	361360, 368040, 376072	359.9 MB (0.351 GiB)
s3_rna_splice_parity_n3	51.8 GiB HG002 RNA-seq (-i splice)	8	VarDictJava	23.21, 23.03, 23.70	23.31 (23.03-23.70)	5324544, 4538880, 5217280	4909.1 MB (4.794 GiB)
s4_perf_budget_4targets_n3	Performance Budget amplicon-tier2	8	VarDictRust / VarDictJava / vardictcpp	Rust=0.01, Java=3.07, C++=5.24	Rust 0.01s	Rust=12032, Java=122964, C++=122572	Rust 11.8 MB / Java 120.1 MB
s4_perf_budget_4targets_n3	Performance Budget hg002-tier2	8	VarDictRust / VarDictJava / vardictcpp	Rust=21.29, Java=153.25, C++=20.10	Rust 21.29s	Rust=487080, Java=10761680, C++=881460	Rust 475.7 MB / Java 10509.5 MB
s4_perf_budget_4targets_n3	Performance Budget hg008-tier2	8	VarDictRust / VarDictJava / vardictcpp	Rust=17.11, Java=77.28, C++=16.91	Rust 17.11s	Rust=528268, Java=17734080, C++=582644	Rust 515.9 MB / Java 17318.4 MB
s4_perf_budget_4targets_n3	Performance Budget seqc2-exome-tier2	8	VarDictRust / VarDictJava / vardictcpp	Rust=277.52, Java=4045.00, C++=287.70	Rust 277.52s	Rust=529060, Java=16471012, C++=2299560	Rust 516.7 MB / Java 16085.0 MB
s4_scale_exome_3way_th1	SEQC2 Whole Exome	1	VarDictRust	805.74, 806.69, 825.81	812.75 (805.74-825.81)	351724, 351844, 351652	343.5 MB (0.335 GiB)
s4_scale_exome_3way_th1	SEQC2 Whole Exome	1	VarDictJava	5767.00, 5789.00, 5717.00	5757.67 (5717.00-5789.00)	1857212, 1887008, 1903816	1838.5 MB (1.795 GiB)
s4_scale_exome_3way_th1	SEQC2 Whole Exome	1	vardictcpp	993.97, 1010.80, 1012.37	1005.71 (993.97-1012.37)	1264172, 1263912, 1264168	1234.5 MB (1.206 GiB)
s4_scale_exome_3way_th2	SEQC2 Whole Exome	2	VarDictRust	405.60, 411.13, 410.70	409.14 (405.60-411.13)	428676, 428772, 417680	415.1 MB (0.405 GiB)

Cell ID	Workload	Threads	Engine	Replicate Wall Times (s)	Wall Mean (Min-Max) (s)	Replicate Peak RSS (KB)	Peak RSS Mean (MB / GiB)
s4_scale_exome_3way_th2	SEQC2 Whole Exome	2	VarDictJava	5762.00, 5839.00, 5806.00	5802.33 (5762.00-5839.00)	5351168, 4222464, 5321168	4848.6 MB (4.735 GiB)
s4_scale_exome_3way_th2	SEQC2 Whole Exome	2	vardictcpp	529.51, 530.65, 531.93	530.70 (529.51-531.93)	1357128, 1357144, 1357020	1325.3 MB (1.294 GiB)
s4_scale_exome_3way_th4	SEQC2 Whole Exome	4	VarDictRust	211.15, 212.32, 213.44	212.30 (211.15-213.44)	462724, 454340, 444064	443.1 MB (0.433 GiB)
s4_scale_exome_3way_th4	SEQC2 Whole Exome	4	VarDictJava	2254.41, 2276.68, 2280.43	2270.51 (2254.41-2280.43)	6932480, 7551728, 6877428	6953.7 MB (6.791 GiB)
s4_scale_exome_3way_th4	SEQC2 Whole Exome	4	vardictcpp	267.51, 267.11, 267.34	267.32 (267.11-267.51)	1438412, 1434796, 1437096	1403.1 MB (1.370 GiB)
s4_scale_exome_3way_th8	SEQC2 Whole Exome	8	VarDictRust	117.36, 119.47, 119.88	118.90 (117.36-119.88)	546844, 553576, 548892	536.9 MB (0.524 GiB)
s4_scale_exome_3way_th8	SEQC2 Whole Exome	8	VarDictJava	1752.96, 1911.57, 1779.37	1814.63 (1752.96-1911.57)	9542144, 9317612, 9601380	9264.7 MB (9.048 GiB)
s4_scale_exome_3way_th8	SEQC2 Whole Exome	8	vardictcpp	138.32, 138.71, 140.40	139.14 (138.32-140.40)	1571092, 1571564, 1557576	1530.0 MB (1.494 GiB)
s4_scale_exome_3way_th16	SEQC2 Whole Exome	16	VarDictRust	75.89, 76.76, 74.66	75.77 (74.66-76.76)	613520, 618784, 633916	607.5 MB (0.593 GiB)
s4_scale_exome_3way_th16	SEQC2 Whole Exome	16	VarDictJava	1885.00, 1929.42, 1866.57	1893.66 (1866.57-1929.42)	18610904, 17986784, 18267580	17859.8 MB (17.441 GiB)
s4_scale_exome_3way_th16	SEQC2 Whole Exome	16	vardictcpp	73.64, 73.74, 75.12	74.17 (73.64-75.12)	1760284, 1774788, 1777884	1729.5 MB (1.689 GiB)
s4_scale_exome_3way_th32	SEQC2 Whole Exome	32	VarDictRust	67.27, 68.20, 68.10	67.86 (67.27-68.20)	763156, 761796, 770404	747.2 MB (0.730 GiB)
s4_scale_exome_3way_th32	SEQC2 Whole Exome	32	VarDictJava	1969.99, 2001.39, 2003.54	1991.64 (1969.99-2003.54)	23221636, 23172288, 22081264	22290.1 MB (21.768 GiB)
s4_scale_exome_3way_th32	SEQC2 Whole Exome	32	vardictcpp	63.41, 64.15, 63.49	63.68 (63.41-64.15)	2151852, 2164356, 2164628	2109.7 MB (2.060 GiB)
s4_span_chr1_3way_th8	Chr1 Span R1000000	8	VarDictRust	1.16, 1.17, 1.20	1.18 (1.16-1.20)	532652, 531732, 531972	519.6 MB (0.507 GiB)
s4_span_chr1_3way_th8	Chr1 Span R1000000	8	vardictcpp	5.50, 5.45, 5.48	5.48 (5.45-5.50)	195336, 195592, 195080	190.8 MB (0.186 GiB)
s4_span_chr1_3way_th8	Chr1 Span R10000000	8	VarDictRust	1.60, 1.60, 1.60	1.60 (1.60-1.60)	1243864, 1184828, 1264216	1202.1 MB (1.174 GiB)
s4_span_chr1_3way_th8	Chr1 Span R10000000	8	vardictcpp	6.73, 6.67, 6.66	6.69 (6.66-6.73)	384668, 386380, 384580	376.2 MB (0.367 GiB)
s4_span_chr1_3way_th8	Chr1 Span R100000000	8	VarDictRust	12.20, 11.98, 11.80	11.99 (11.80-12.20)	1277644, 1245396, 1311332	1248.2 MB (1.219 GiB)
s4_span_chr1_3way_th8	Chr1 Span R100000000	8	vardictcpp	16.45, 16.32, 16.36	16.38 (16.32-16.45)	489044, 456176, 494436	468.6 MB (0.458 GiB)
s4_span_chr1_3way_th8	Chr1 Span R248956422	8	VarDictRust	27.18, 27.21, 27.77	27.39 (27.18-27.77)	1455976, 1458324, 1428280	1413.6 MB (1.380 GiB)
s4_span_chr1_3way_th8	Chr1 Span R248956422	8	vardictcpp	30.39, 30.33, 30.27	30.33 (30.27-30.39)	857536, 814916, 821808	811.9 MB (0.793 GiB)
s4_span_chr1_3way_th16	Chr1 Span R1000000	16	VarDictRust	1.14, 1.12, 1.12	1.13 (1.12-1.14)	532756, 533136, 532956	520.5 MB (0.508 GiB)
s4_span_chr1_3way_th16	Chr1 Span R1000000	16	vardictcpp	5.13, 5.09, 5.16	5.13 (5.09-5.16)	354976, 352696, 354120	345.6 MB (0.338 GiB)
s4_span_chr1_3way_th16	Chr1 Span R10000000	16	VarDictRust	0.94, 0.94, 0.96	0.95 (0.94-0.96)	1799560, 1909712, 1909552	1829.0 MB (1.786 GiB)
s4_span_chr1_3way_th16	Chr1 Span R10000000	16	vardictcpp	5.50, 5.76, 5.83	5.70 (5.50-5.83)	690112, 690496, 694340	675.4 MB (0.660 GiB)
s4_span_chr1_3way_th16	Chr1 Span R100000000	16	VarDictRust	6.23, 6.23, 6.38	6.28 (6.23-6.38)	2036552, 2025764, 2006076	1975.4 MB (1.929 GiB)
s4_span_chr1_3way_th16	Chr1 Span R100000000	16	vardictcpp	10.75, 10.72, 10.56	10.68 (10.56-10.75)	836088, 828196, 814352	806.9 MB (0.788 GiB)
s4_span_chr1_3way_th16	Chr1 Span R248956422	16	VarDictRust	14.60, 15.14, 14.81	14.85 (14.60-15.14)	2439756, 2382796, 2433268	2361.9 MB (2.307 GiB)
s4_span_chr1_3way_th16	Chr1 Span R248956422	16	vardictcpp	17.70, 18.57, 18.90	18.39 (17.70-18.90)	1209508, 1203464, 1169436	1166.2 MB (1.139 GiB)
s4_span_chr1_3way_th32	Chr1 Span R1000000	32	VarDictRust	1.17, 1.17, 1.16	1.17 (1.16-1.17)	535504, 533984, 535260	522.4 MB (0.510 GiB)
s4_span_chr1_3way_th32	Chr1 Span R1000000	32	vardictcpp	5.39, 5.42, 5.55	5.45 (5.39-5.55)	399284, 397120, 400828	389.7 MB (0.381 GiB)
s4_span_chr1_3way_th32	Chr1 Span R10000000	32	VarDictRust	0.91, 0.92, 0.91	0.91 (0.91-0.92)	2788972, 2733968, 2726516	2685.4 MB (2.622 GiB)

Cell ID	Workload	Threads	Engine	Replicate Wall Times (s)	Wall Mean (Min-Max) (s)	Replicate Peak RSS (KB)	Peak RSS Mean (MB / GiB)
s4_span_chr1_3way_th32	Chr1 Span R10000000	32	vardictcpp	6.04, 6.10, 6.16	6.10 (6.04-6.16)	1262680, 1249556, 1240044	1221.5 MB (1.193 GiB)
s4_span_chr1_3way_th32	Chr1 Span R100000000	32	VarDictRust	5.34, 5.38, 5.16	5.29 (5.16-5.38)	2946828, 3016304, 3043928	2932.0 MB (2.863 GiB)
s4_span_chr1_3way_th32	Chr1 Span R100000000	32	vardictcpp	10.79, 10.30, 10.00	10.36 (10.00-10.79)	1443032, 1493356, 1510604	1447.6 MB (1.414 GiB)
s4_span_chr1_3way_th32	Chr1 Span R248956422	32	VarDictRust	11.90, 11.96, 11.76	11.87 (11.76-11.96)	3828452, 3939028, 3823184	3773.0 MB (3.685 GiB)
s4_span_chr1_3way_th32	Chr1 Span R248956422	32	vardictcpp	16.17, 16.25, 16.26	16.23 (16.17-16.26)	1821828, 1762144, 1756708	1738.5 MB (1.698 GiB)
s4_span_chr1_f0	Chr1 10 Mbp All-Positions (-f 0)	8	VarDictRust	20.09, 19.41, 19.24	19.58 (19.24-20.09)	6185864, 6185476, 6185552	6040.6 MB (5.899 GiB)
s4_span_chr1_f0	Chr1 10 Mbp All-Positions (-f 0)	8	vardictcpp	6.04, 5.99, 5.99	6.01 (5.99-6.04)	386612, 380596, 386420	375.5 MB (0.367 GiB)
s4_jobs_per_node_2x4c_n3	Concurrent Job 1 (4c Exome)	4	VarDictRust	212.56, 209.64, 211.94	211.38 (209.64-212.56)	449244, 446504, 448288	437.5 MB (0.427 GiB)
s4_jobs_per_node_2x4c_n3	Concurrent Job 2 (4c Exome)	4	VarDictRust	212.43, 209.86, 210.85	211.05 (209.86-212.43)	453764, 451428, 452540	442.0 MB (0.432 GiB)
s5_wgs_hg002_full_contigs_3way_th32	HG002 Full WGS (22 Unchunked Contigs)	32	VarDictRust	1516.61	1516.61 (25:16.61)	4928384	4.93 GB (4.70 GiB)
s5_wgs_hg002_full_giab_3way_th32	HG002 Full WGS (≤50 kbp GIAB BED)	32	VarDictRust	1221.21	1221.21 (20:21.21)	1433128	1.43 GB (1.37 GiB)
s5_wgs_hg008_full_contigs_3way_th32	HG008 Full WGS (22 Unchunked Contigs, -U)	32	VarDictRust	10192.00	10192.00 (2:49:52)	12135724	12.14 GB (11.57 GiB)
s5_wgs_hg008_full_giab_3way_th32	HG008 Full WGS (≤50 kbp GIAB BED, -U)	32	VarDictRust	6355.00	6355.00 (1:45:55)	4877136	4.88 GB (4.65 GiB)

All n=3 runs represent technical replicates on identical GCP Intel Xeon Platinum 8373C Ice Lake instances.

Supplementary Table S4 (Table S4). Verified Truth-Set Operating Points, Variant-Class Stratification & 18-Arm Somatic Noise-Floor Matrix

Source Cell ID	Cohort / Platform	Stratum / Regime / Operating Point	Engine / Parity	Primary Metric 1 (TP / Candidates / SNV)	Primary Metric 2 (FP / PASS / InDel)	Summary Score / F1
s4_tier3_truth_and_qual_sweep	HG002 GRCh38	All-PASS (Unfiltered)	VarDictRust == VarDictJava	TP=23,062	FP=22,458, FN=813	P=0.5038, R=0.9659, F1=0.6622
s4_tier3_truth_and_qual_sweep	HG002 GRCh38	Optimal QUAL (≥128.000)	VarDictRust == VarDictJava	TP=23,004	FP=553, FN=871	P=0.9763, R=0.9635, F1=0.9699
s4_tier3_truth_and_qual_sweep	SEQC2 HCC1395	All-PASS (Unfiltered)	VarDictRust == VarDictJava	TP=1,069	FP=11,334, FN=140	P=0.0852, R=0.8842, F1=0.1555
s4_tier3_truth_and_qual_sweep	SEQC2 HCC1395	Optimal QUAL (≥126.000)	VarDictRust == VarDictJava	TP=794	FP=293, FN=415	P=0.7279, R=0.6567, F1=0.6905
s4_tier3_truth_and_qual_sweep	HG008 GIAB v0.3	All-PASS (Unfiltered)	VarDictRust == VarDictJava	TP=619	FP=1,565, FN=288	P=0.2687, R=0.6825, F1=0.3856
s4_tier3_truth_and_qual_sweep	HG008 GIAB v0.3	Optimal QUAL (≥176.000)	VarDictRust == VarDictJava	TP=593	FP=38, FN=314	P=0.9354, R=0.6538, F1=0.7696
evidence_certified	HG005 GRCh38	All-PASS (Unfiltered)	VarDictRust == VarDictJava	TP=29	FP=2, FN=1	P=0.9355, R=0.9667, F1=0.9508
evidence_certified	HG005 GRCh38	Optimal QUAL (≥226.0)	VarDictRust == VarDictJava	TP=29	FP=1, FN=1	P=0.9667, R=0.9667, F1=0.9667
evidence_certified	NA24385 GRCh37	All-PASS (Unfiltered)	VarDictRust == VarDictJava	TP=5	FP=81, FN=0	P=0.0581, R=1.0000, F1=0.1099
evidence_certified	NA24385 GRCh37	Optimal QUAL (≥160.0)	VarDictRust == VarDictJava	TP=5	FP=0, FN=0	P=1.0000, R=1.0000, F1=1.0000
s4_tier3_truth_and_qual_sweep	HG002	Variant-Class & AF Breakdown	rust	SNV=209,030, Ins=2,305, Del=10,975	MNP=2,065, SV=3,187	PASS=48,447 / Total=227,562
s4_tier3_truth_and_qual_sweep	HG002	Variant-Class & AF Breakdown	java	SNV=209,030, Ins=2,305, Del=10,975	MNP=2,065, SV=3,187	PASS=48,447 / Total=227,562
s4_tier3_truth_and_qual_sweep	HG002	Variant-Class & AF Breakdown	cpp	SNV=209,066, Ins=2,309, Del=10,949	MNP=2,049, SV=3,187	PASS=48,431 / Total=227,560
s4_tier3_truth_and_qual_sweep	SEQC2	Variant-Class & AF Breakdown	rust	SNV=594,344, Ins=8,966, Del=15,716	MNP=4,383, SV=35	PASS=120,307 / Total=623,444
s4_tier3_truth_and_qual_sweep	SEQC2	Variant-Class & AF Breakdown	java	SNV=594,344, Ins=8,966, Del=15,716	MNP=4,383, SV=35	PASS=120,307 / Total=623,444
s4_tier3_truth_and_qual_sweep	SEQC2	Variant-Class & AF Breakdown	cpp	SNV=594,344, Ins=8,966, Del=15,715	MNP=4,383, SV=34	PASS=120,305 / Total=623,442
s4_tier3_truth_and_qual_sweep	HG008	Variant-Class & AF Breakdown	rust	SNV=15,213, Ins=1,161, Del=1,410	MNP=614, SV=28	PASS=11,358 / Total=18,426

Source Cell ID	Cohort / Platform	Stratum / Regime / Operating Point	Engine / Parity	Primary Metric 1 (TP / Candidates / SNV)	Primary Metric 2 (FP / PASS / InDel)	Summary Score / F1
s4_tier3_truth_and_qual_sweep	HG008	Variant-Class & AF Breakdown	java	SNV=15,213, Ins=1,161, Del=1,410	MNP=614, SV=28	PASS=11,358 / Total=18,426
s4_tier3_truth_and_qual_sweep	HG008	Variant-Class & AF Breakdown	cpp	SNV=15,214, Ins=1,161, Del=1,408	MNP=614, SV=28	PASS=11,356 / Total=18,425
s5_cross_platform_short_read_chr20_panel	Element AVITI 111x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=61	PASS Somatic=13	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Element AVITI 111x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=61	PASS Somatic=13	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Element AVITI 111x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (Java == Rust)	Total Candidates=186	PASS Somatic=13	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	PacBio Onso 136x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=76	PASS Somatic=19	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	PacBio Onso 136x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=78	PASS Somatic=19	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	PacBio Onso 136x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (Java == Rust)	Total Candidates=185	PASS Somatic=19	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Illumina 195x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=115	PASS Somatic=24	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Illumina 195x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=131	PASS Somatic=24	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Illumina 195x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (Java == Rust)	Total Candidates=1,199	PASS Somatic=23	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Ultima ppmSeq 157x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=574	PASS Somatic=133	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Ultima ppmSeq 157x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=581	PASS Somatic=136	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Ultima ppmSeq 157x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (Java == Rust)	Total Candidates=2,791	PASS Somatic=159	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Roche SBX 119x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=770	PASS Somatic=293	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Roche SBX 119x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=788	PASS Somatic=295	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Roche SBX 119x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (Java == Rust)	Total Candidates=3,354	PASS Somatic=296	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Ultima UG100 306x	Min VAF 1%, ≥2 Reads (-f 0.01 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=944	PASS Somatic=376	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Ultima UG100 306x	All VAFs, ≥2 Reads (-f 0 -r 2)	100% Exact Match (Java == Rust)	Total Candidates=1,519	PASS Somatic=472	0 diffs vs. VarDictJava (eps=1e-6)
s5_cross_platform_short_read_chr20_panel	Ultima UG100 306x	Singleton Noise Floor (-f 0 -r 1)	100% Exact Match (Java == Rust)	Total Candidates=4,994	PASS Somatic=538	0 diffs vs. VarDictJava (eps=1e-6)
s2_fig2_hg002_chr20_truth	HG002 Chr20 Exome (Daniels et al. 2024)	7-Caller Truth Benchmark	VarDictRust == VarDictJava	Calls=7,422	SNV F1=0.9858	InDel F1=0.8956 (see Table 3)
s5_grch38_field_callers_truth	HG002, SEQC2 & HG008 GRCh38	17-Run Native GRCh38 Caller Matrix	VarDictRust == VarDictJava	HG002 SNV F1=0.6619	HG002 InDel F1=0.7177	100% Exact Match (see Table 3)
s6_low_vaf_titration_truth_curve	HG008 Tumor-Purity Titration (20% to 1%)	30-Run Low-VAF Somatic Curve (-f 0.002)	VarDictRust == VarDictJava	20% Purity SNV F1=0.7512	1% Purity SNV F1=0.0569	100% Exact Match across all 5 purities (see Table 3)
s6_hg002_v50q_cmrg_mosaic_eval	HG002 v4.2.1, v5.0q T2T-Q100, CMRG & Mosaic	12-Evaluation Next-Gen Germline & Mosaic Suite	VarDictRust == VarDictJava (≤50 kbp BED)	v5.0q SNV F1=0.6650	CMRG SNV F1=0.6791	Mosaic Recall=98.82% (see Table 7)
s6_hg008_modern_ml_and_commercial_callers	HG008 GIAB v0.4 Somatic Benchmark	11-Caller Modern ML & Commercial Landscape	VarDictRust v1.0.6 Full WGS & Slice	Full WGS SNV F1=0.7886	Full WGS InDel F1=0.4898	25 evaluations across 11 callers (see Table 7)

VarDictRust and VarDictJava produce identical rtg vcfeval accuracy at both All-PASS and Optimal QUAL operating points across all 5 truth cohorts and 18/18 exact-match noise-floor regimes.

Supplementary Table S5 (Table S5). Complete 70-Cell GA4GH v3.5 Genomic Stratification Matrix (7 Strata x 10 Platforms)

Cell ID	Cohort	Sequencing Platform	GA4GH v3.5 Stratum	SNV Precision	SNV Recall	SNV F1	InDel Precision / Recall / F1
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	Homopolymer (4–6 bp)	0.4869	0.9645	0.6471	0.4874 / 0.9134 / 0.6356
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	Homopolymer (7–11 bp)	0.6056	0.9278	0.7328	0.6079 / 0.9474 / 0.7406
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	Homopolymer (≥12 bp)	0.6828	0.8704	0.7653	0.8869 / 0.7491 / 0.8122
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	Tandem Repeats	0.5462	0.9126	0.6833	0.8782 / 0.8357 / 0.8564
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	Segmental Duplications	0.5343	0.8980	0.6700	0.5650 / 0.8514 / 0.6792
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	Low Mappability	0.5202	0.8309	0.6398	0.4419 / 0.8199 / 0.5742
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Illumina	RefSeq CDS	0.3644	0.9831	0.5317	0.1209 / 1.0000 / 0.2157
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	Homopolymer (4–6 bp)	0.9491	0.9685	0.9587	0.9371 / 0.9308 / 0.9339

Cell ID	Cohort	Sequencing Platform	GA4GH v3.5 Stratum	SNV Precision	SNV Recall	SNV F1	InDel Precision / Recall / F1
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	Homopolymer (7–11 bp)	0.9657	0.9325	0.9488	0.9626 / 0.9533 / 0.9579
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	Homopolymer (≥12 bp)	0.9793	0.8765	0.9251	0.9882 / 0.8968 / 0.9403
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	Tandem Repeats	0.8662	0.9383	0.9008	0.9430 / 0.8396 / 0.8883
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	Segmental Duplications	0.7358	0.8980	0.8088	0.7485 / 0.8446 / 0.7937
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	Low Mappability	0.6786	0.8419	0.7515	0.6406 / 0.7640 / 0.6969
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Element AVITI	RefSeq CDS	0.9311	0.9887	0.9591	0.7857 / 1.0000 / 0.8800
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	Homopolymer (4–6 bp)	0.5416	0.9632	0.6934	0.4645 / 0.9279 / 0.6191
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	Homopolymer (7–11 bp)	0.5844	0.9284	0.7173	0.2319 / 0.9519 / 0.3729
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	Homopolymer (≥12 bp)	0.4531	0.8796	0.5981	0.7337 / 0.8764 / 0.7987
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	Tandem Repeats	0.4990	0.9326	0.6502	0.6606 / 0.8336 / 0.7371
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	Segmental Duplications	0.5267	0.8960	0.6634	0.4792 / 0.8581 / 0.6150
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	Low Mappability	0.3691	0.8269	0.5104	0.3342 / 0.7888 / 0.4695
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 MGI DNBSEQ	RefSeq CDS	0.4110	0.9831	0.5796	0.2703 / 0.9091 / 0.4167
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	Homopolymer (4–6 bp)	0.8208	0.9409	0.8768	0.0195 / 0.7546 / 0.0381
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	Homopolymer (7–11 bp)	0.5572	0.7097	0.6243	0.0514 / 0.4792 / 0.0928
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	Homopolymer (≥12 bp)	0.2666	0.6944	0.3853	0.2957 / 0.4573 / 0.3591
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	Tandem Repeats	0.7497	0.9100	0.8221	0.2424 / 0.7895 / 0.3709
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	Segmental Duplications	0.8726	0.8755	0.8740	0.0277 / 0.7297 / 0.0535
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	Low Mappability	0.8497	0.7799	0.8133	0.0359 / 0.6708 / 0.0682
s6_ga4gh_v35_stratification_matrix	HG002 Germline	HG002 Ultima UG100	RefSeq CDS	0.7489	0.9568	0.8402	0.0016 / 0.7273 / 0.0033
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	Homopolymer (4–6 bp)	0.9545	0.8750	0.9130	0.7778 / 0.4118 / 0.5385
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	Homopolymer (7–11 bp)	1.0000	1.0000	1.0000	1.0000 / 0.7647 / 0.8667
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	Homopolymer (≥12 bp)	0.8000	0.5000	0.6154	0.8235 / 0.3373 / 0.4786
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	Tandem Repeats	0.1296	0.6364	0.2154	0.0563 / 0.2105 / 0.0889
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	Segmental Duplications	0.9000	0.7500	0.8182	0.8333 / 0.5556 / 0.6667
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	Low Mappability	0.3158	1.0000	0.4800	0.3000 / 1.0000 / 0.4615
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Illumina	RefSeq CDS	1.0000	1.0000	1.0000	No Truth Indels in Stratum
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	Homopolymer (4–6 bp)	0.8958	0.8958	0.8958	0.9375 / 0.8824 / 0.9091
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	Homopolymer (7–11 bp)	1.0000	1.0000	1.0000	1.0000 / 0.8824 / 0.9375
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	Homopolymer (≥12 bp)	0.8000	0.5000	0.6154	0.9500 / 0.6867 / 0.7972
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	Tandem Repeats	0.0909	0.6364	0.1591	0.1154 / 0.3158 / 0.1690
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	Segmental Duplications	0.7143	0.8333	0.7692	0.8333 / 0.5556 / 0.6667
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	Low Mappability	0.2683	0.9167	0.4151	0.2727 / 1.0000 / 0.4286
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Element AVITI	RefSeq CDS	1.0000	1.0000	1.0000	No Truth Indels in Stratum
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	Homopolymer (4–6 bp)	0.9286	0.8125	0.8667	0.6667 / 0.7059 / 0.6857
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	Homopolymer (7–11 bp)	0.7500	0.7500	0.7500	0.8125 / 0.7647 / 0.7879
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	Homopolymer (≥12 bp)	0.8000	0.5000	0.6154	0.9231 / 0.5783 / 0.7111
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	Tandem Repeats	0.0976	0.7273	0.1720	0.0625 / 0.1579 / 0.0896
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	Segmental Duplications	0.6000	0.7500	0.6667	0.6667 / 0.4444 / 0.5333
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	Low Mappability	0.2353	1.0000	0.3810	0.2143 / 1.0000 / 0.3529
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 PacBio Onso	RefSeq CDS	1.0000	1.0000	1.0000	No Truth Indels in Stratum
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	Homopolymer (4–6 bp)	0.8571	0.8750	0.8660	0.4000 / 0.2353 / 0.2963
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	Homopolymer (7–11 bp)	0.8000	1.0000	0.8889	0.7000 / 0.4118 / 0.5185
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	Homopolymer (≥12 bp)	1.0000	0.5000	0.6667	0.3913 / 0.1084 / 0.1698
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	Tandem Repeats	0.7143	0.4545	0.5556	0.5000 / 0.1053 / 0.1739
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	Segmental Duplications	0.8889	0.6667	0.7619	0.7500 / 0.3333 / 0.4615
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	Low Mappability	0.7857	0.9167	0.8462	0.5000 / 1.0000 / 0.6667
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Roche SBX	RefSeq CDS	1.0000	1.0000	1.0000	No Truth Indels in Stratum
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	Homopolymer (4–6 bp)	0.8667	0.8125	0.8387	0.2857 / 0.1176 / 0.1667
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	Homopolymer (7–11 bp)	1.0000	0.2500	0.4000	0.0000 / 0.0000 / 0.0000
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	Homopolymer (≥12 bp)	1.0000	0.1250	0.2222	1.0000 / 0.0241 / 0.0471
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	Tandem Repeats	0.2500	0.6364	0.3590	0.0435 / 0.0526 / 0.0476
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	Segmental Duplications	0.7273	0.6667	0.6957	0.5000 / 0.2222 / 0.3077
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	Low Mappability	0.6154	0.6667	0.6400	1.0000 / 0.3333 / 0.5000
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima ppmSeq	RefSeq CDS	1.0000	1.0000	1.0000	No Truth Indels in Stratum
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	Homopolymer (4–6 bp)	0.9512	0.8125	0.8764	0.5000 / 0.1176 / 0.1905

Cell ID	Cohort	Sequencing Platform	GA4GH v3.5 Stratum	SNV Precision	SNV Recall	SNV F1	InDel Precision / Recall / F1
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	Homopolymer (7–11 bp)	1.0000	0.2500	0.4000	0.0000 / 0.0000 / 0.0000
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	Homopolymer (≥12 bp)	1.0000	0.1250	0.2222	0.2308 / 0.0361 / 0.0625
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	Tandem Repeats	0.6250	0.4545	0.5263	0.1429 / 0.0526 / 0.0769
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	Segmental Duplications	1.0000	0.8333	0.9091	1.0000 / 0.2222 / 0.3636
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	Low Mappability	0.9167	0.9167	0.9167	0.5000 / 0.3333 / 0.4000
s6_ga4gh_v35_stratification_matrix	HG008 Somatic	HG008 Ultima UG100	RefSeq CDS	1.0000	1.0000	1.0000	No Truth Indels in Stratum

All 70 stratification cells confirm 0 site differences between VarDictRust v1.0.6 and VarDictJava v1.8.3.

Supplementary Table S6 (Table S6). 13-Mode Functional Parity Matrix and CPP-1..10 vardictcpp Divergence Taxonomy

Source Cell ID	Mode / Divergence ID	VarDictJava Rows / Behavior	VarDictRust Rows / Behavior	vardictcpp Rows / Behavior	Verification Verdict & Root Cause
s2_tierp_mode_parity_sweep	simple_f0.01	228	228 (FP=0, FN=0)	228 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	simple_f0_mrd	248	248 (FP=0, FN=0)	248 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	simple_chunk50000	228	228 (FP=0, FN=0)	228 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	simple_fisher	650	650 (FP=0, FN=0)	650 (FP=1, FN=1)	MATCH
s2_tierp_mode_parity_sweep	targeted_panel_chr17	83	83 (FP=0, FN=0)	83 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	sv_deletion_regions	3	3 (FP=0, FN=0)	3 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	low_coverage	146	146 (FP=0, FN=0)	146 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	amplicon_c17amp	145	145 (FP=0, FN=0)	145 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	amplicon_fisher	145	145 (FP=0, FN=0)	145 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	somatic_f0.01_big	664	664 (FP=0, FN=0)	664 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	somatic_f0_mrd_c17	1396	1,396 (FP=0, FN=0)	1,396 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	somatic_fisher_c17	527	527 (FP=0, FN=0)	527 (FP=0, FN=0)	MATCH
s2_tierp_mode_parity_sweep	P5_cram_twin	0_NoClassDefFoundError	228 (FP=0, FN=0)	228 (FP=0, FN=0)	MATCH_BAM_TWIN
s5_h4_maintenance_security_matrix	CPP-1: VCF Header & INFO/END Schema	Standard VCF 4.2 + END tag	Exact v1.8.3 header & END parity	Malformed/omitted END & header lines	Schema mismatch in downstream parsers
s5_h4_maintenance_security_matrix	CPP-2: CRISPR (-J/-j) & Structural Modes	Supported	Supported (100% MATCH)	Unimplemented / drops flags	Verified on extended-coverage-tier2
s5_h4_maintenance_security_matrix	CPP-3: Amplicon (-a) Output Format	38/40-col .var + VCF	Exact 38/40-col .var + VCF MATCH	Emits raw TSV instead of VCF	Breaks compare_vcf.py & bcftools
s5_h4_maintenance_security_matrix	CPP-4: Concatenated CLI Flag Parsing	Supported	Supported (60-flag parity)	Exits with code 120 on bundled flags	CLI wrapper incompatibility
s5_h4_maintenance_security_matrix	CPP-5: Chunk Boundary Deduplication (--chunk)	Single-region (or OOM)	100 kbp halo tiling + cross-tile dedup	Emits +309,569 duplicate boundary calls on unchunked HG002 WGS	Uncoordinated 50 kbp chunk splitting
s5_h4_maintenance_security_matrix	CPP-6: Missing ##contig Header Lines	Emits ##contig from .fai	Emits exact ##contig lines from .fai	Omits ##contig header lines	Requires header injection for rtg vcfeval
s5_h4_maintenance_security_matrix	CPP-7: GRCh37 / b37 Chromosome Naming	Supported	Supported (NA24385-b37 EQUAL)	Hardcoded chr-prefix assumptions	Tested on na24385-b37-truth
s5_h4_maintenance_security_matrix	CPP-8: BAM/CRAM Reference & Decode	BAM (CRAM needs classpath)	Native BAM + CRAM (-G) zero-diff	Fails or diverges on CRAM twins	Verified on hg008-cram-twin & Ultima CRAM
s5_h4_maintenance_security_matrix	CPP-9: MNP Splitting & Somatic SV Drops	Reference pileup & realigner	Exact parity with VarDictJava	Differs on 394 callable HG002 WGS calls & drops SVs	Realignment & tie-break divergence
s5_h4_maintenance_security_matrix	CPP-10: RNA-Seq Splice Mode (-i)	Supported (12,668 rows on HG002 mRNA)	Supported (12,668 rows 100% MATCH)	Unimplemented (SKIP)	Verified on 51.8 GiB HG002 STAR BAM

Every divergence class was empirically observed and cataloged during the 60-cell GCP evaluation.